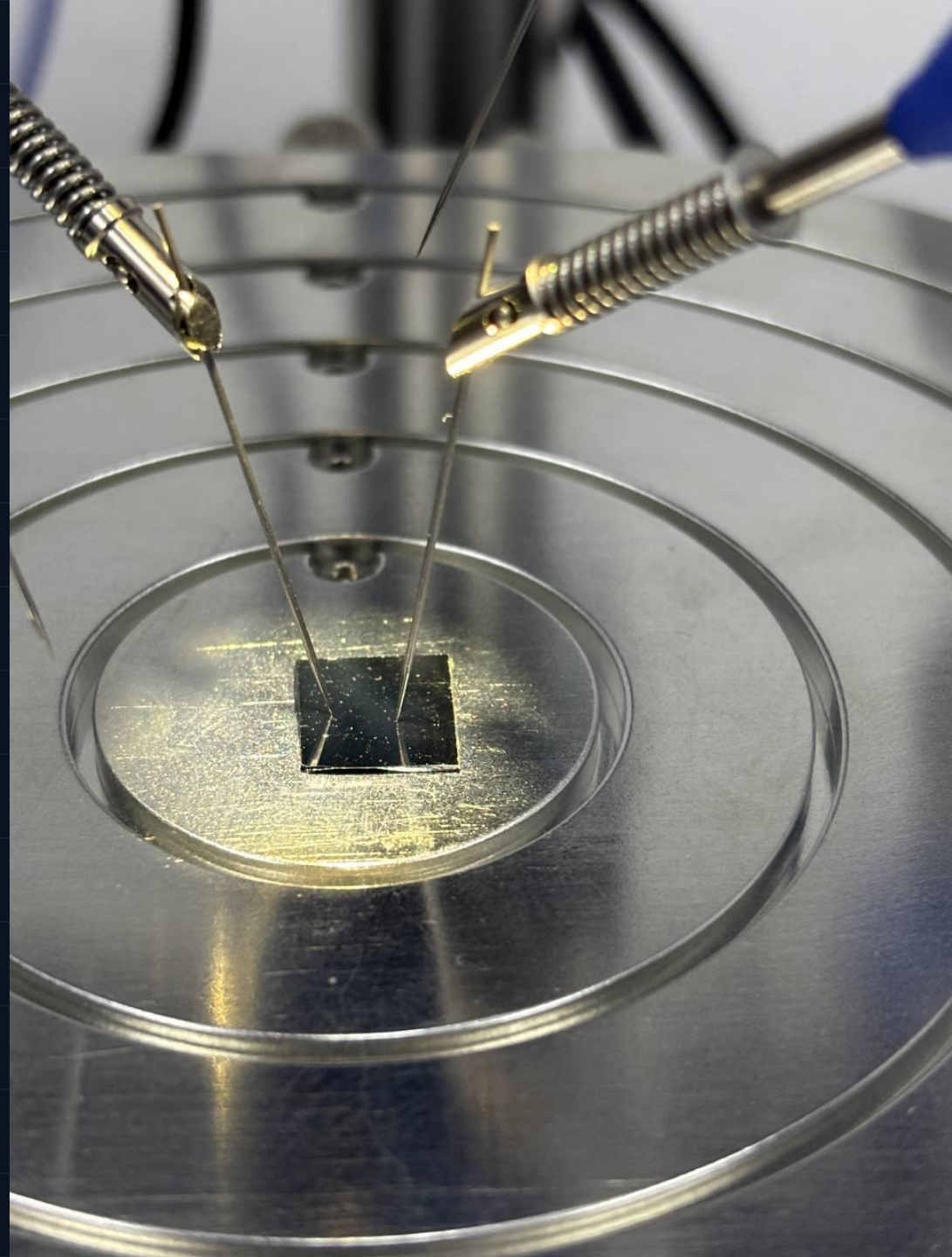
Autonomous AI-Driven Physics Laboratory

A closed-loop AI agent that controls an Ossila SMU, analyzes data in real time and writes the report, demonstrated on 5 nm Ag/Si thin films.

LangGraph · FastAPI · Streamlit · Ossila SMU



From manual measurements to an autonomous lab

The problem

- Nanoscale film (5 nm Ag/Si) measurements are done by hand
- Engineer picks voltage ranges and runs every sweep
- Data processed separately in Python / Origin
- Formulas applied and the report written manually

Our solution

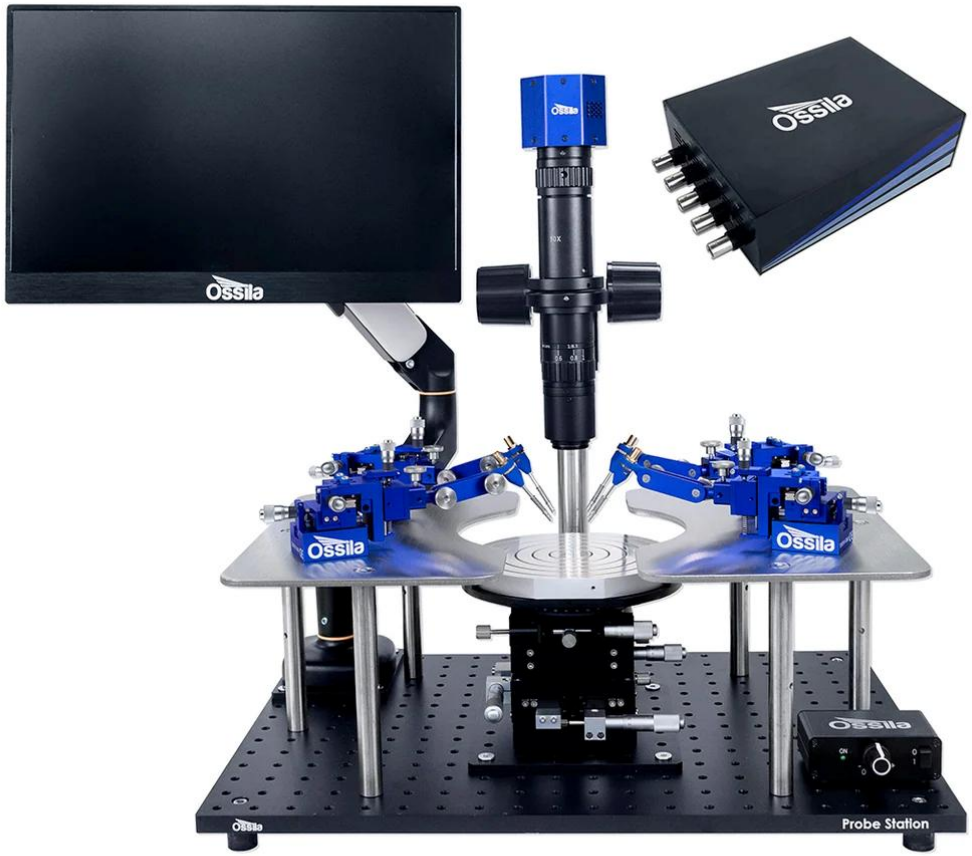
- AI agent acts as an autonomous physicist-experimentalist
- Drives the Ossila SMU and reads results live
- Decides the next measurement step by itself
- Computes physical parameters and generates a PDF report

System overview: one click, five layers



Results flow back the same way: measured data → physics summary → plots → PDF → live dashboard

The instrument: Ossila Source Measure Unit



A compact laboratory instrument that combines a precise DC voltage source with a voltmeter and an ammeter. It applies a voltage and simultaneously measures the resulting current, which is the basis of I–V characterization of thin films, solar cells, LEDs and transistors.

±10 V
source and measure range

±200 mA
maximum current

5
current ranges, 10 nA–200 mA

2 + 2
SMU + VSense channels

USB / LAN
PC control, text commands

BNC
measurement connections

Hardware layer: Ossila SMU on COM3

COM3	115200	5 nm	≈ 4 Ω
USB-UART bridge	baud rate	Ag on Si/SiO_2	sample resistance

Built-in hardware protection (core/ossila_connector.py)

Voltage clamping ±1.2 V

A 4 Ω sample would draw huge current at ±5 V and trip the SMU current compliance. The software limits voltage automatically.

Relay arc protection

Mechanical relays switch with a 50–150 ms pause, preventing contact arcing and instrument lock-ups.

Singleton connector

get_shared_connector() opens COM3 only once, avoiding Windows “Access is denied” errors between FastAPI and the agent.

The AI brain: LangGraph agent



Primary brain

Groq Cloud API

ChatGroq with openai/gpt-oss-120b for fast, capable tool-calling decisions.

Five instruments (tools) the agent can call

1. measure_2point_resistance_tool
2. sweep_voltage_tool
3. adaptive_fine_sweep_tool
4. hysteresis_sweep_tool
5. measure_time_series_it_tool

Five-step closed-loop measurement cycle

1

Baseline check

0.5 V applied; initial ohmic resistance $R_0 \approx 4.16\ \Omega$

2

Linear I–V sweep

−1.0 V to +1.0 V, 15 points; dI/dV and V_critical

3

Adaptive fine scan

Agent centers a dense scan around V_critical

4

Hysteresis loop

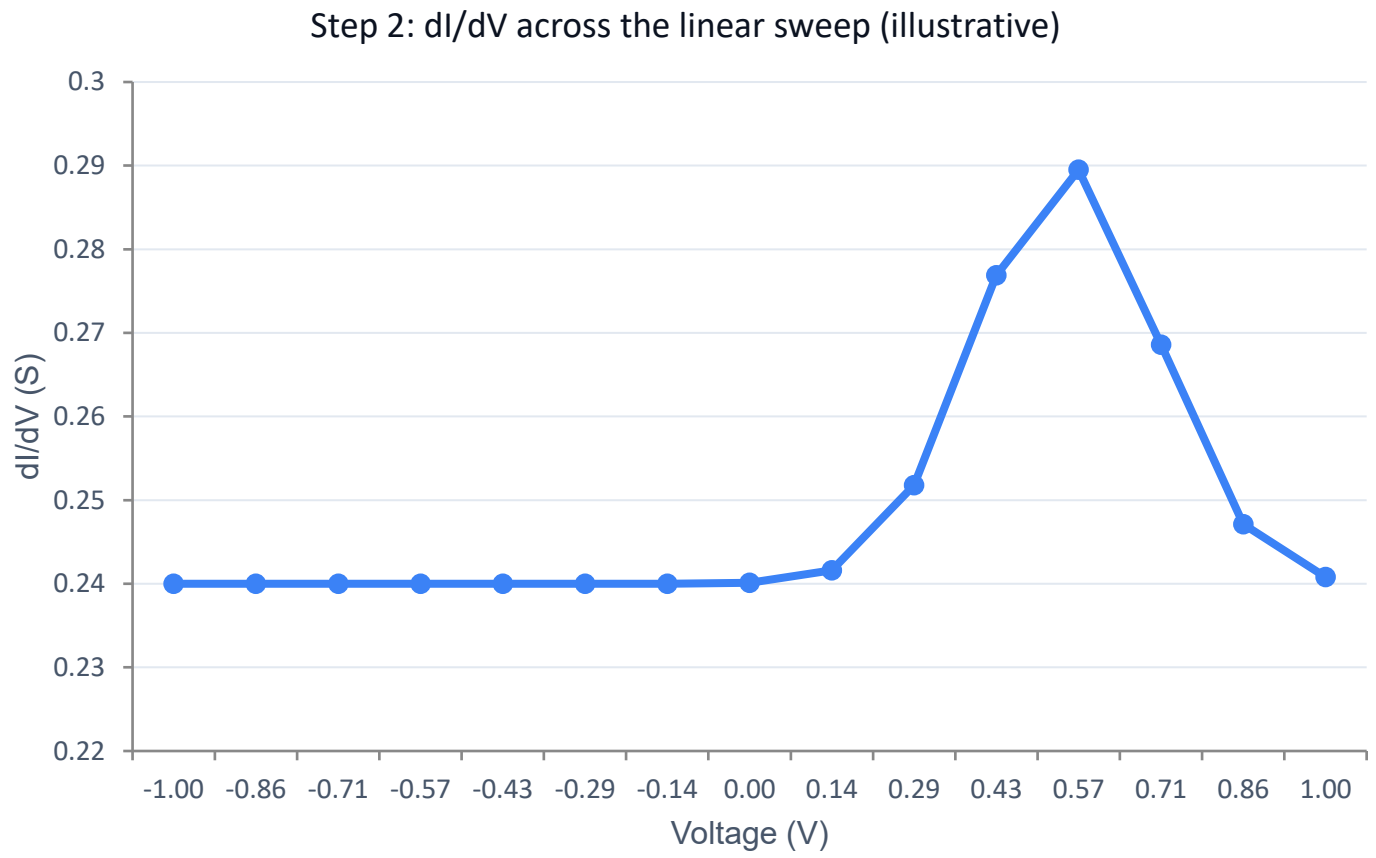
Bipolar cyclic sweep ±1.0 V for memristive loop area

5

Time series I(t)

0.5 V for 5 s; current dynamics for 1/f noise PSD

Closed-loop AI



Read

Agent reads the Step 2 result and finds V_critical, the transition voltage.

Decide

It chooses, by itself, the center of a dense measurement window.

Act

adaptive_fine_sweep_tool runs a fine scan around V_critical.

Illustrative curve for explanation – replace with your measured dI/dV if desired.

Physics engine: five parameters from raw data

Ohmic resistance R_0

$$R_0 = V / I$$

Low-voltage slope of the I–V curve.

SCLC exponent n

$$I \propto V^n, \quad n = d(\ln I) / d(\ln V)$$

n > 1.5: conduction dominated by trap filling.

Fowler–Nordheim barrier Φb

$$\ln(I / V^2) \propto -1 / V$$

Slope gives the effective Schottky/FN barrier height in eV.

Hysteresis area

$$\text{Area} = \oint I(V)\, dV$$

Measure of charge trapping and energy loss (memristive).

1/f noise PSD

Welch method on I(t)

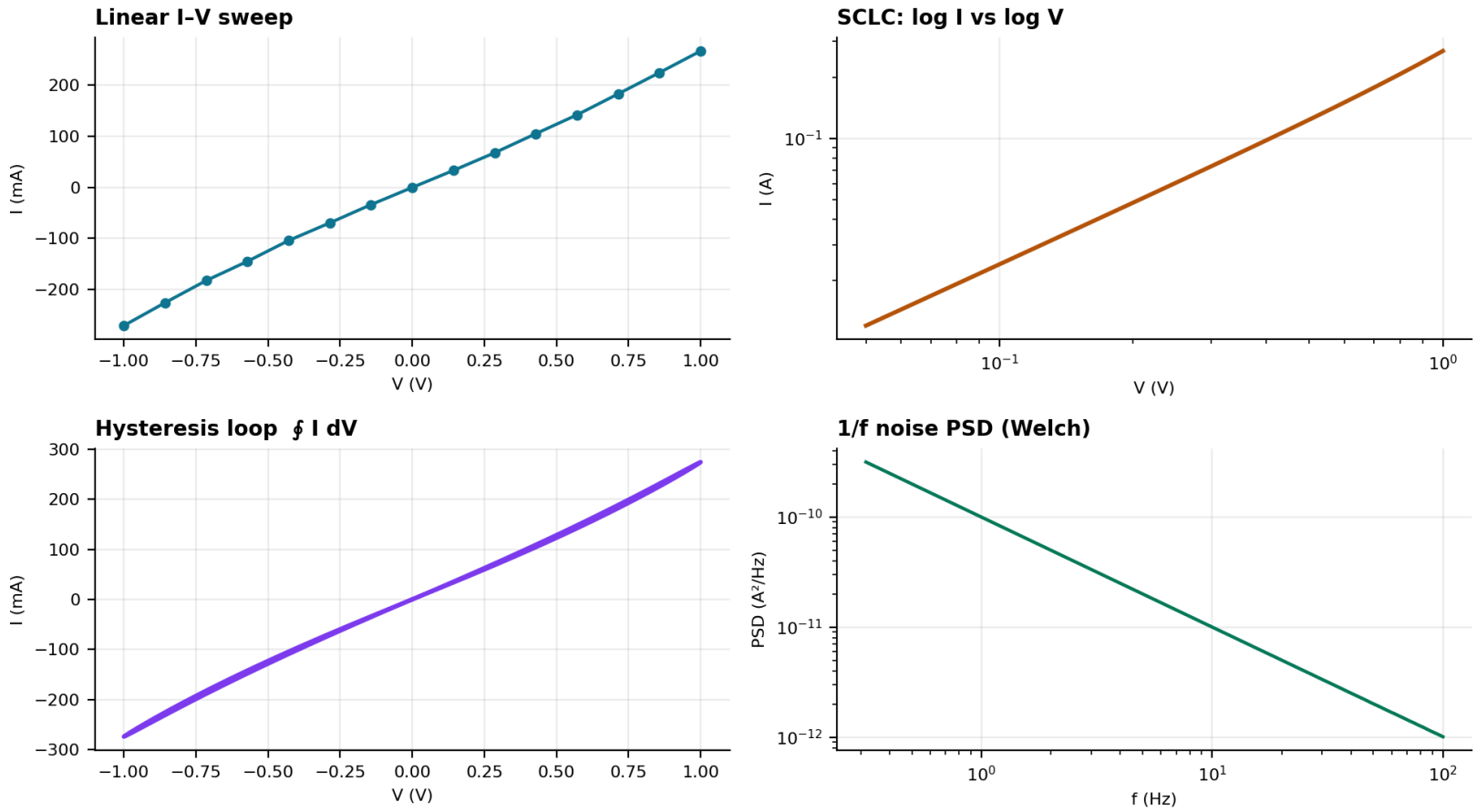
Mean noise level in A^2/Hz.

Saved output

`physics_summary.json`

Feeds the plots, the PDF report and the dashboard tables.

Automatic 300 DPI four-panel result figure



I–V sweep

coarse + adaptive points

SCLC / FN

log-log and ln(I/V²) analysis

Hysteresis loop

loop area $\oint I\,dV$

1/f noise

Welch PSD from I(t)

Illustrative layout generated for this deck. Replace with data/plots_output.png from your real run.

Software architecture

File	Technology	Role
`main.py`	Python subprocess	Master orchestrator: starts FastAPI (8000) and Streamlit (8501) with one command
`backend/main.py`	FastAPI, Uvicorn	Async REST API between UI and hardware; keeps the UI from freezing
`frontend/app.py`	Streamlit, Plotly	5-tab dark glassmorphic dashboard with interactive plots and PDF download
`core/ossila_connector.py`	PySerial	Direct COM3 driver: voltage clamp, relay pauses, line-buffered readline()
`core/agent_graph.py`	LangGraph, LangChain	Dual LLM/SLM agent with 5 tools running the closed-loop cycle
`core/physics_analysis.py`	NumPy, SciPy	Computes R_o, n, Φb, loop area and 1/f PSD from JSON data
`services/generate_plots.py`	Matplotlib	300 DPI four-panel figure (data/plots_output.png)
`services/generate_report.py`	ReportLab	PDF with AI log, physics tables and plots (data/report.pdf)

The dashboard: everything in one view



- Interactive Plotly graphs
- Ossila device specs
- Physics tables
- One-click PDF download

UI mockup for illustration. Replace with a real screenshot of the Streamlit app.

End-to-end flow in about 20 seconds

1 — 2 — 3 — 4 — 5 — 6 — 7

Launch

py main.py starts FastAPI + Streamlit

Click

Start Full Characterization

API call

/api/v1/agent/run starts the agent in a background thread

Measure

Agent calls the 5 tools; raw data saved as JSON

Analyze

SCLC, FN, hysteresis and 1/f computed and saved

Report

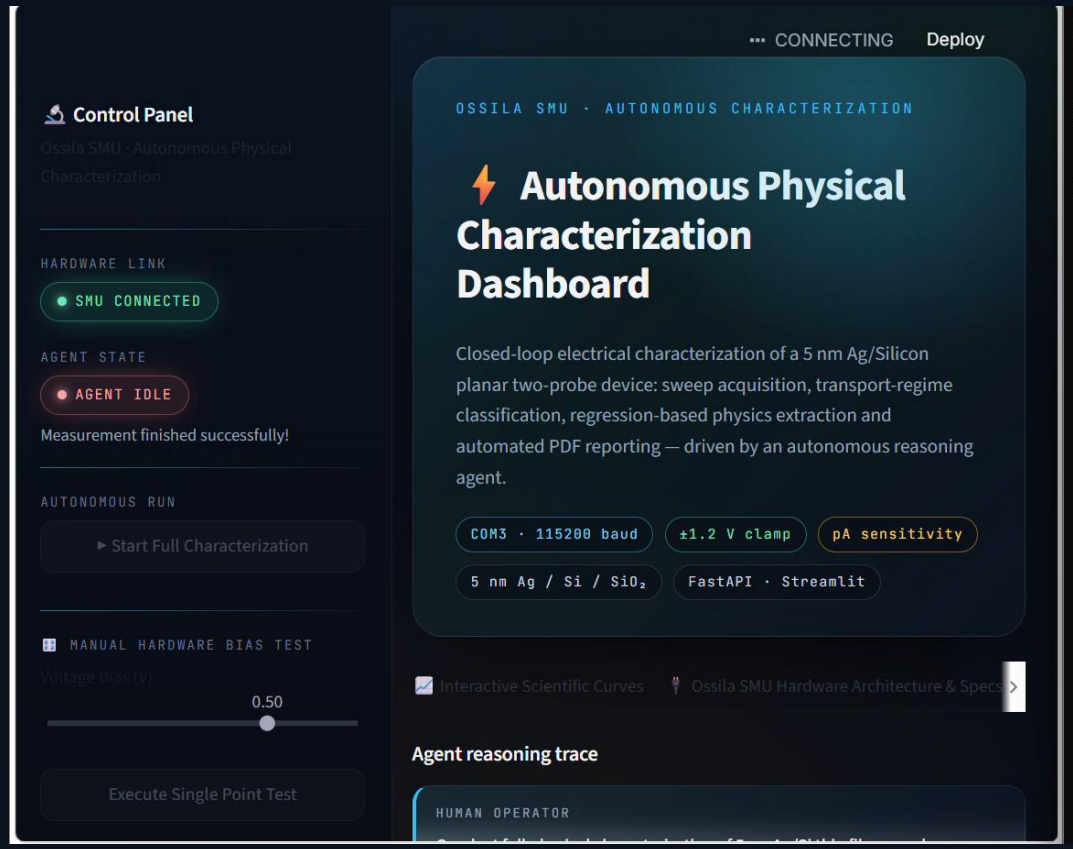
Plots and data/report.pdf generated

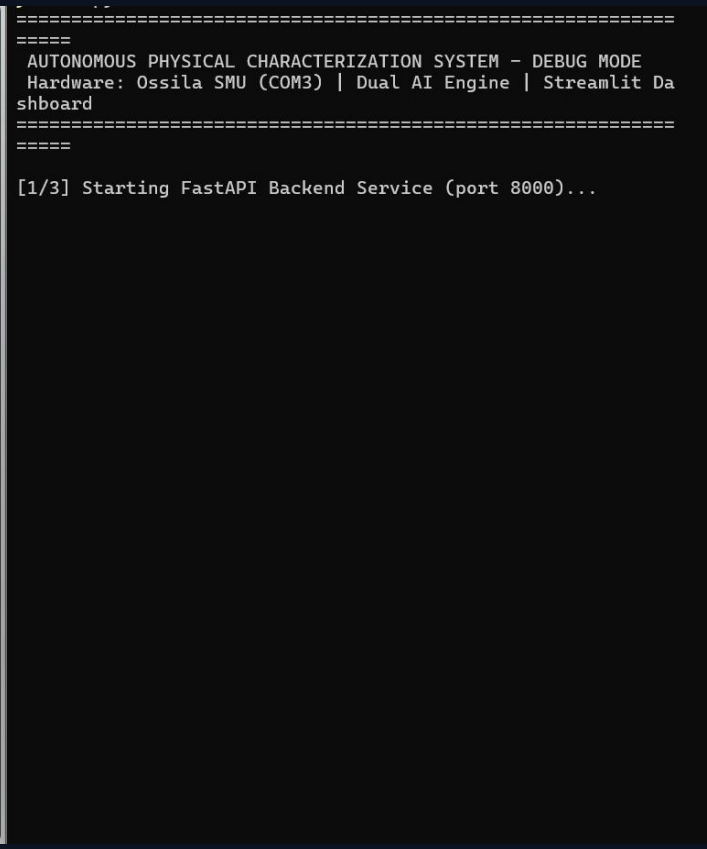
Refresh

Dashboard updates; PDF button activates

Live demo




CONNECTING
Deploy
Control Panel
HARDWARE LINK
SMU CONNECTED
AGENT STATE
AGENT IDLE
Measurement finished successfully!
AUTONOMOUS RUN
Start Full Characterization
MANUAL HARDWARE BIAS TEST
0.50
Execute Single Point Test
OSSILA SMU · AUTONOMOUS CHARACTERIZATION
Autonomous Physical Characterization Dashboard
Closed-loop electrical characterization of a 5 nm Ag/Silicon planar two-probe device: sweep acquisition, transport-regime classification, regression-based physics extraction and automated PDF reporting — driven by an autonomous reasoning agent.
COM3 · 115200 baud
±1.2 V clamp
pA sensitivity
5 nm Ag / Si / SiO₂
FastAPI · Streamlit
Interactive Scientific Curves
Ossila SMU Hardware Architecture & Specs
Agent reasoning trace
HUMAN OPERATOR


AUTONOMOUS PHYSICAL CHARACTERIZATION SYSTEM – DEBUG MODE
Hardware: Ossila SMU (COM3) | Dual AI Engine | Streamlit Dashboard
[1/3] Starting FastAPI Backend Service (port 8000)...

Three key strengths

Hardware protection

Voltage clamping ±1.2 V, relay delays and compliance protection keep the sample and the Ossila SMU safe.

Closed-loop intelligence

The agent reads the first sweep and dynamically decides the center of the adaptive fine scan; it does not just run code.

Full automation

About 10 seconds after the button press, a print-quality PDF report is ready.

Thank you

Questions & discussion

Autonomous AI-Driven Physics Laboratory · 5 nm Ag/Si characterization

